Garuda Ramkiran

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Education

Vellore Institute of Technology (VIT), Vellore 2024 – 2028
B.Tech, Computer Science and Engineering | CGPA: 9.28 / 10 *Vellore, India*
Coursework: Computer Networks, Operating Systems, DBMS, Compiler Design, Data Structures & Algorithms, AWS Cloud
Class XII, New Baldwins International: 90%, JEE Main 96th percentile (2024) | Class X, Narayana Public School: 95.2% (2022)

Security Projects

SentinelMesh | *C++, ESP32, ML-KEM, ML-DSA, AES-GCM, Python* **1st Place, Code Cortex 3.0 (TAM VIT)**
- Post-quantum secured radio mesh for battery-powered sensors; team of 4, research paper in progress. I built the handshake firmware, the attack experiments and the malware/phishing detectors.
- Wrote a 3-message key exchange (ML-KEM-768 + ML-DSA-65, then AES-256-GCM) over ESP-NOW. A handshake does not fit a 250-byte frame, so it is sent as about 39 fragments; median setup 439 ms on real boards.
- The attack: with no keys you cannot break the crypto, but each forged HELLO forces a signature check that burns battery. With no defence, 584 of 606 forged HELLOs in a 2-minute flood reached full verification.
- Added a check that runs *before* verification (packet rate, failures, RSSI; a 423-byte decision tree in C) plus a stateless cookie challenge. Version 1 also blocked the real gateway (0 of 10) because forged and real packets shared a sender ID; version 2 challenges instead of dropping: about 2 of 601 forged HELLOs got through, 9 of 9 real handshakes succeeded.
- Insider with a valid key: 5 of 566 requests got through in the first 1.5 s before the gate reacted, and one that answers the cookie would bypass it; both written up as open gaps.
- Fixed a replay-protection bug found on hardware: after a reboot the gateway picked a lower random epoch and the sensor rejected every HELLO as a replay. Physical tamper (light, reed switch, motion, voltage) wipes session keys.
- Attacked our own detectors. Malware model (LightGBM on 54 PE-header fields, 138k files): copying 14 header fields from benign programs dropped detection from 20/20 to 4/20; it also flagged 40.5% of modern pip/npm binaries (0.44% after retraining). Phishing model: padding emails with normal text dropped recall from 99.3% to 15.8%; adversarial training recovered 79.1%.

Backstop – Guardrails for a Payment Agent | *Python, FastAPI, Razorpay API* Razorpay Buildathon
- The agent proposes payment retries and messages; a separate rules layer (card-network retry caps, RBI notice, quiet hours) can veto but never widen an action, and every decision goes to an append-only audit log.
- Tested idempotency on the live API instead of trusting docs: `order.create`'s `receipt` is not an idempotency key (two identical calls made two orders).

GridForge – Running Untrusted Code | *Python, FastAPI, Redis, Docker*
- Runs user-uploaded Python projects in Docker containers with networking disabled, a read-only mount, a non-root user, 512 MB / 1 CPU / 100-process limits and a timeout.

Flash-Sale Checkout Engine | *Python, FastAPI, PostgreSQL, Redis, k6*
- Fault-injection testing found a logic bug: a retried request confirmed the same order twice; fixed with idempotency keys. Zero oversells under load across 3 locking strategies.

Experience

iFocus Systec India Pvt Ltd May 2026 – Jul 2026
Software Developer Intern *Bangalore*
- Built TicketZilla, the company's IT helpdesk, used daily by 300+ employees (React, NestJS, PostgreSQL, Redis), with role-based access for 7 roles.
- Compared 9 hosted language models for QwikHire's job-description generator on accuracy, output format and how easily user input could hijack the prompt; shipped GPT-OSS 120B/20B with a fallback.

ElectraCode May 2026 – Jul 2026
Contract Developer *Bangalore*
- Built a school ERP sized for 850 schools / 255,000 students (FastAPI, PostgreSQL). Enforced per-school isolation so staff can only read or change their own school's records; fixed audit-log and stock-transfer bugs.

SRS Learning Center (SRS EduTech) May 2025 – Present
Operations & Technology *Bengaluru*
- Built StaffRoom, a teacher-hiring platform (Next.js, PostgreSQL, NextAuth); run franchising and business development.

Technical Skills

Security: post-quantum crypto (ML-KEM, ML-DSA), AES-GCM, replay protection, flood/DoS defence, attacking ML detectors, static PE analysis (pefile), access control, container sandboxing
Languages: C++, C, Python, Go, TypeScript, Java, SQL
Embedded: ESP32, PlatformIO, ESP-NOW, I2C sensors
Backend & Tools: FastAPI, NestJS, PostgreSQL, Redis, Docker, Linux, Git, k6, Prometheus, Grafana

Achievements & Leadership

- 1st Place, Code Cortex 3.0 (TAM VIT, security track) | 1st Prize, Startup Street 9.0 Ideathon | Finalist, DevSpace 54-Hour Hackathon
- Treasurer, Toastmasters club | Senior Core, ACM VIT Student Chapter | Ultramarathon runner (50 km)